Winter Storm Fern Situation Report

January 2026

Formatted based on LA Fire Situation Report

Summary

A major American winter storm, often referred to as Winter Storm Fern, started on Friday, January 23rd 2026 and continued through January 26th 2026. The storm brought heavy snow and freezing rain to several U.S. states, ranging from the southern plains to the Eastern United States. Specifically, Tennessee, Kentucky Several states experienced network outages and extremely cold temperatures.

The following visualizations are developed based on Meta AI for Good's data through Facebook. The data covers network coverage and population density from January 27 to February 10, with a focus on Tennessee and Kentucky and daily updates during the collection period.

How the Data Is Constructed

The Facebook data used in this report comes from Meta's Data for Good crisis datasets. In plain terms, these datasets use activity from Facebook users or Facebook business Pages to estimate how people, population density, network coverage, and business activity change during a crisis. The data does not represent everyone in the affected area. For population and movement datasets, it only represents Facebook app users who have Location Services enabled. For business activity, it represents qualifying Facebook business Pages with enough activity to be included while preserving privacy.

The data is constructed by comparing what is observed during the crisis period to what was typical before the crisis. For example, population density during the storm is compared to a pre-crisis baseline. Movement during the storm is compared to normal movement patterns between the same places before the event. Business activity is compared to normal posting behavior from business pages before the crisis. Because the data is relative to a baseline, the main signal is not the raw count itself, but whether a place is above or below its expected level.

Geographically, the data is reported in two main ways: Bing tiles and administrative regions. Bing tiles divide space into small grid cells. At Bing tile level 14, each tile is roughly 2.4 kilometers on a side near the equator, or about the size of several city blocks. Administrative regions instead use boundaries such as counties, states, or other official geographic areas.

Temporally, the population and movement datasets use fixed 8-hour windows. These windows begin at 00:00, 08:00, and 16:00 Pacific Time. This means that the time periods do not automatically adjust to local time zones. For Tennessee and Kentucky, the reported time windows are still based on Pacific Time, which should be kept in mind when interpreting daily patterns. Business activity is reported daily, based on the date value in the dataset. Some Meta crisis datasets are provided directly as Bing tile data, while others are provided as administrative-region data. When the data is provided as Bing tiles, each observation is tied to a grid cell with a latitude and longitude. When we add county information to those data, we are not changing the original measurement. Instead, we are assigning each tile or point to the county that contains it. This allows the tile-level data to be summarized at the county level.

For example, a Bing tile located inside Davidson County, Tennessee would be assigned the Davidson County name and GEOID. If many Bing tiles fall within Davidson County, their values can then be averaged, summed, or otherwise summarized to produce a county-level measure. This is useful for reporting, but it is not identical to receiving an administrative-region dataset directly from Meta. The administrative-region dataset is already aggregated within county or state boundaries by Meta, while the Bing tile version requires us to overlay the tile locations onto county boundaries.

Facebook Data Definitions

- **Baseline Ratio**: Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the baseline period over users in all tiles in the boundary box defining the crisis
- **Crisis Ratio**: Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the 8-hour window over users in all tiles in the boundary box defining the crisis
- **Population Change Density (%)**: Percent change in Facebook Location Services-enabled user population counts per tile, calculated by dividing the difference by the baseline (plus a small value, usually 1)
- **Time Windows**: aggregated over 3 fixed 8-hour time windows, beginning at 00:00, 08:00 and 16:00, that represent changes in population density

- **Bing Tiles**: equivalent to roughly 2.4 kilometers on a side near the equator or the size of 6 x 6 city blocks
- **Administrative Regions**: the polygon boundaries for administrative regions derived from the Database of Global Administrative Areas, also known as GADM.

Outside Data Added

In addition to the Meta crisis datasets, county-level demographic data was added from the American Community Survey. This outside data provides context about the counties affected by the storm, including total population, median household income, poverty rate, share of residents age 65 and older, and share of households without vehicle access. These variables help interpret which communities may be more vulnerable during a crisis.

The demographic data is joined using `county_geoid`, a standardized county identifier. This is important because county names alone are not unique across states. For example, many states have counties with the same name, but each county has a unique GEOID.

Aggregation Choices

For county-level maps, the data is summarized by county and date. When multiple observations fall within the same county on the same date, the report uses the average percent change to represent the county. This choice is appropriate for percent-based indicators because summing percentages would not be meaningful.

For population count variables, such as baseline population or crisis population, summing can be used when the goal is to estimate total observed users across tiles or regions. For percent change, z-scores, or ratios, averaging is more appropriate because these variables already describe relative change.

For movement data, origin and destination counties are treated separately. Each row represents movement from a starting location to an ending location, so the analysis keeps both `start_county_geoid` and `end_county_geoid`. This avoids incorrectly collapsing movement into a single county and preserves the direction of movement.

Limitations

These datasets should be interpreted as indicators of crisis-related change, not exact measures of the full population. The population and movement data only include Facebook users with Location Services enabled, so the sample may not represent all residents equally. Areas with fewer Facebook users or lower Location Services usage may be less reliable.

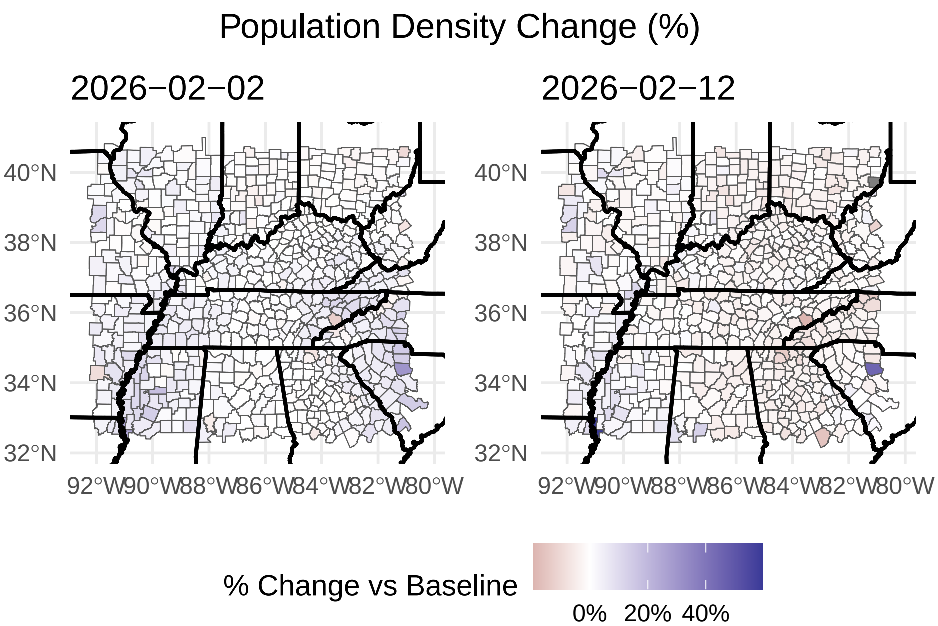
Privacy protections also affect interpretation. Small counts may be suppressed or removed, which can especially affect rural areas. For movement data, this means it is not reliable to calculate exact net flows by simply adding inbound and outbound movement, because some small movement vectors are excluded.

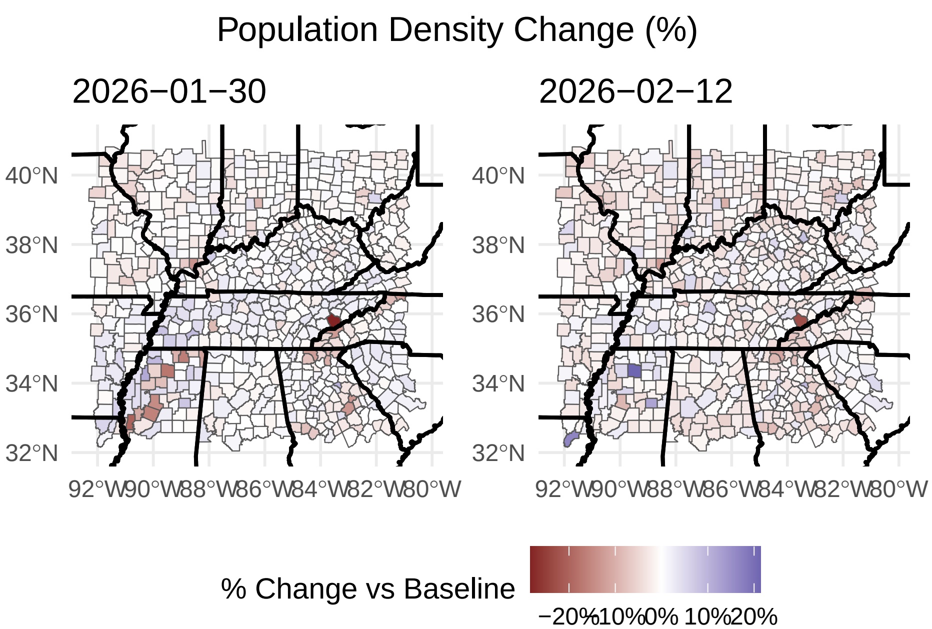
The baseline period is limited to the pre-crisis period available to Meta. This means the analysis does not have a long historical record of pre-crisis behavior. As a result, seasonal changes, holidays, or unusual pre-crisis conditions may affect the baseline. The data is useful for rapid situational awareness, but it should not be interpreted as a complete long-term population or economic dataset.

Finally, the data is time-limited. The documentation notes that some datasets are accessible for 90 days after the final update day and then discarded, which limits longer-term reproducibility and longitudinal analysis.

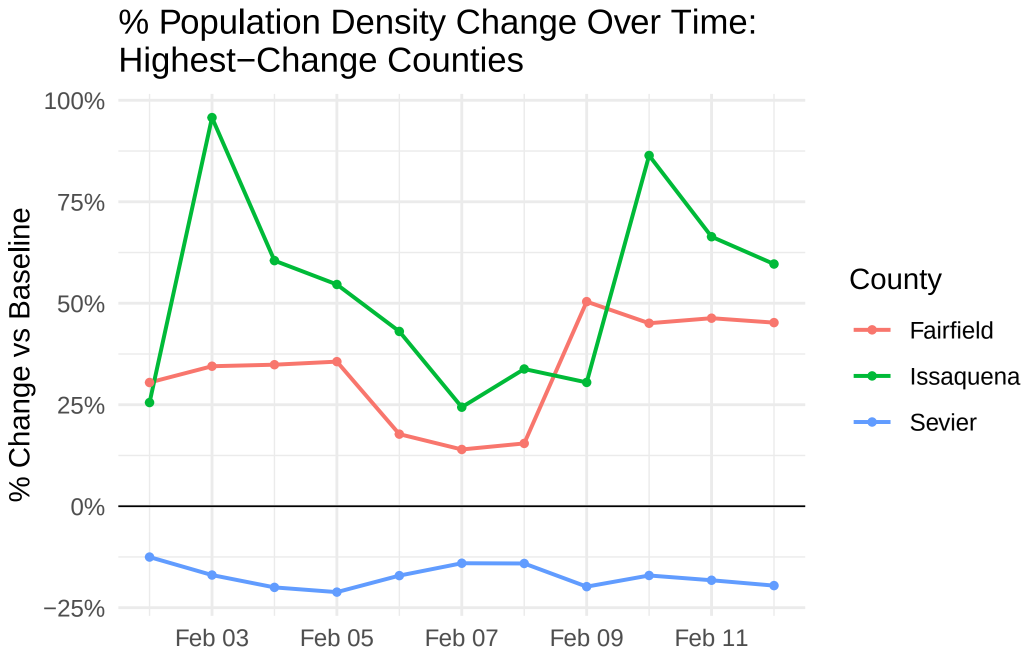
Movement

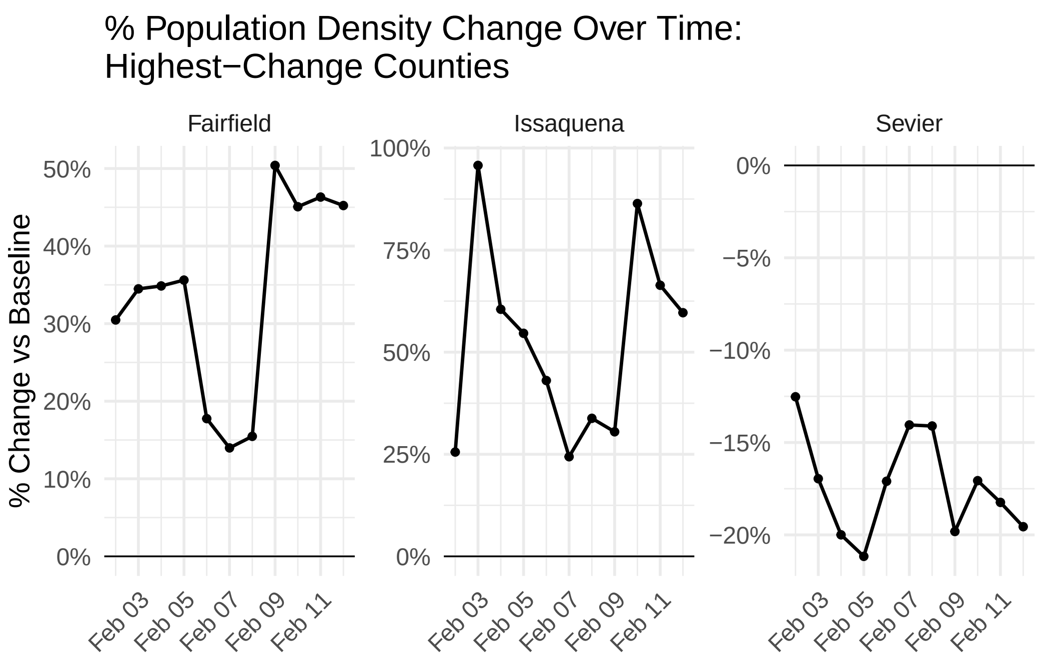
% Population Density Change

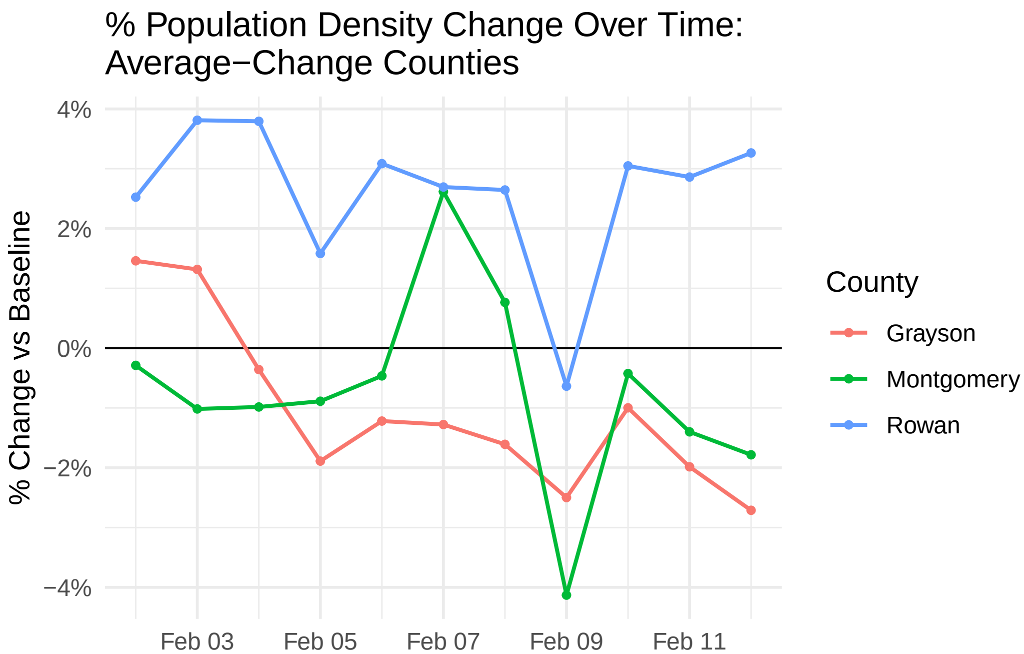


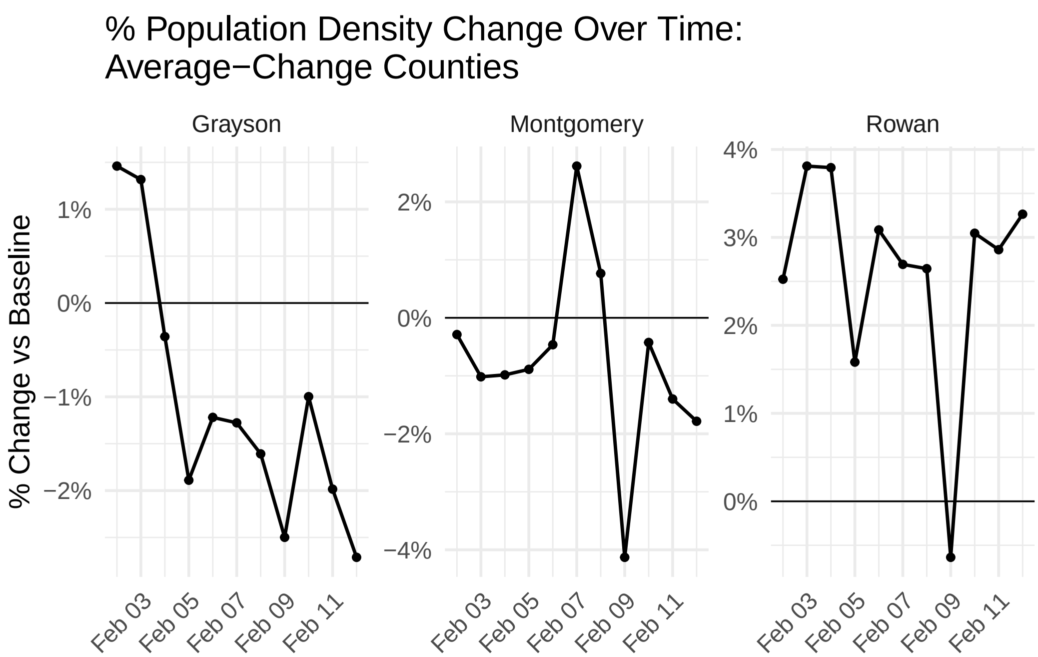


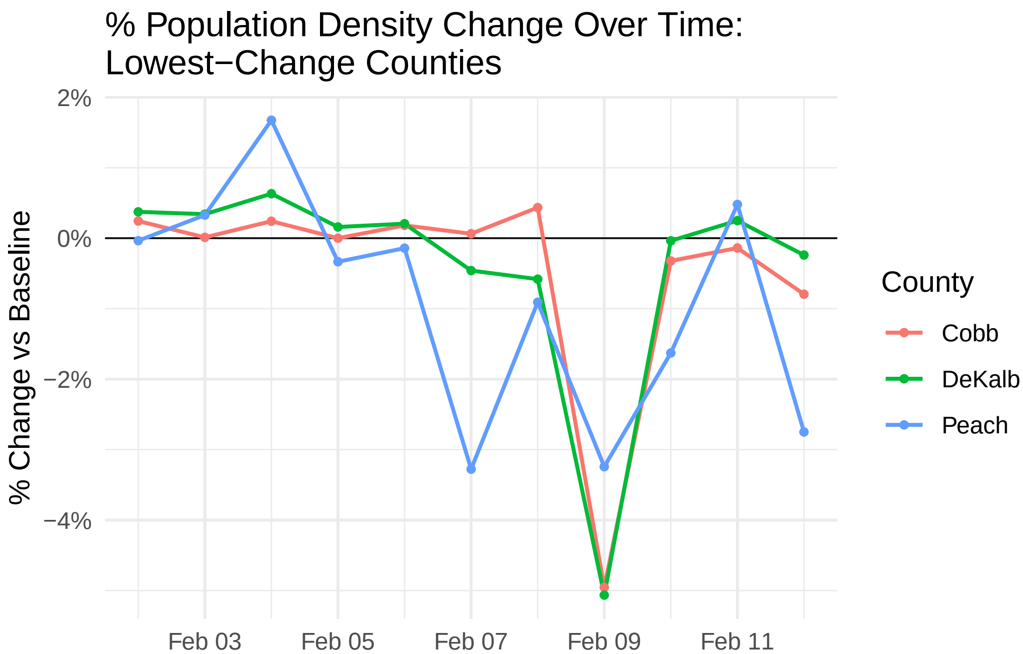
Time Series of Movement

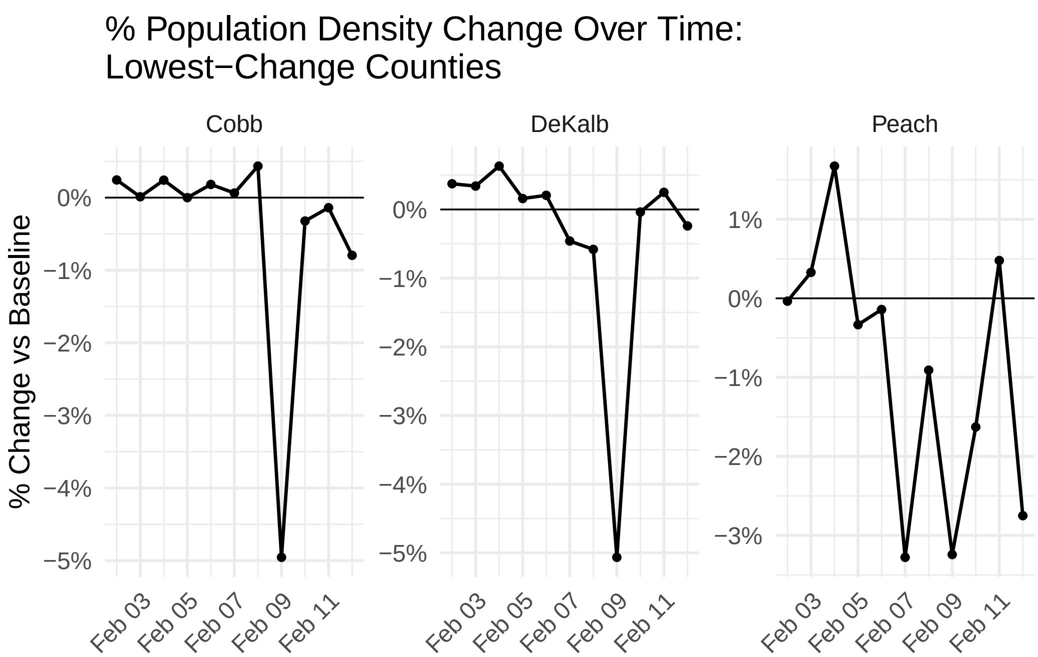





% Population Density Change Over Time:
Average−Change Counties
% Change vs Baseline
4%
2%
0%
−2%
−4%
Feb 03
Feb 05
Feb 07
Feb 09
Feb 11
County
Grayson
Montgomery
Rowan


% Population Density Change Over Time:
Average−Change Counties
Grayson
Montgomery
Rowan
% Change vs Baseline
1%
0%
−1%
−2%
2%
0%
−2%
−4%
4%
3%
2%
1%
0%
Feb 03
Feb 05
Feb 07
Feb 09
Feb 11


% Population Density Change Over Time:
Lowest−Change Counties
% Change vs Baseline
2%
0%
−2%
−4%
Feb 03
Feb 05
Feb 07
Feb 09
Feb 11
County
Cobb
DeKalb
Peach


% Population Density Change Over Time:
Lowest−Change Counties
Cobb
DeKalb
Peach
% Change vs Baseline
0%
−1%
−2%
−3%
−4%
−5%
1%
Feb 03
Feb 05
Feb 07
Feb 09
Feb 11

Places

Table 1

start_county_name	end_county_name	total_crisis_movement	avg_z_score
Jefferson	Jefferson	9345516	0.232645726
Franklin	Franklin	7508643	-0.161025222
Hamilton	Hamilton	6839250	-0.174952705
Madison	Madison	6210069	0.031685405
Shelby	Shelby	6061752	-0.104994055
Marion	Marion	5944828	-0.152500253
St. Louis	St. Louis	5402795	-0.190837292
Montgomery	Montgomery	4517362	-0.157323653
Fulton	Fulton	3727658	-0.269724915
Gwinnett	Gwinnett	3714756	-0.502521525
Davidson	Davidson	3594106	0.590149281
Knox	Knox	3197785	0.448540936
Fayette	Fayette	2946948	0.007203512
Cobb	Cobb	2784631	-0.036566617
Greenville	Greenville	2755429	0.555334176
DeKalb	DeKalb	2717337	-0.302742152
Warren	Warren	2276887	-0.268528368
Butler	Butler	2142991	-0.038897247
Rutherford	Rutherford	2139230	-0.049582873
Richland	Richland	2081487	0.658837921

Counties with Most Increased Population Density on 2026-02-02

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Fairfield	South Carolina	30.5	20,550	$47,885	15.6%	3.2%	5.9%
Issaquena	Mississippi	25.5	928	$31,429	16.2%	2.7%	14.0%
Grenada	Mississippi	17.9	21,217	$48,804	24.6%	2.3%	7.9%
Hampton	South Carolina	16.6	18,254	$44,711	20.7%	2.2%	11.0%
York	South Carolina	14.5	293,673	$89,095	8.4%	2.2%	3.7%
Orangeburg	South Carolina	13.8	83,253	$45,675	22.4%	3.1%	9.0%
Chester	South Carolina	13.8	32,182	$52,458	18.7%	2.5%	6.3%
Holmes	Mississippi	13.5	16,191	$32,538	36.4%	1.9%	15.9%
Lincoln	North Carolina	12.7	92,716	$80,016	11.2%	2.9%	3.3%
Lake	Tennessee	12.5	6,579	$28,814	32.2%	2.2%	18.3%
Alexander	North Carolina	12.2	36,412	$65,354	12.6%	2.4%	3.1%
Quitman	Mississippi	11.0	5,774	$32,412	29.6%	2.4%	10.9%
Alexander	Illinois	10.7	4,875	$47,043	19.0%	4.4%	10.1%
Gaston	North Carolina	10.7	234,881	$67,478	13.5%	2.3%	5.1%
Osage	Missouri	10.6	13,402	$76,681	8.6%	2.6%	5.6%

Counties with Most Decreased Population Density on 2026-02-02

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-12.5	99,652	$62,581	13.5%	2.7%	4.1%
Carroll	Ohio	-9.6	26,659	$64,835	12.9%	3.5%	6.8%
Jefferson	Arkansas	-8.9	64,802	$51,096	18.6%	2.8%	7.5%
Clay	West Virginia	-6.7	7,835	$42,318	26.4%	3.5%	10.3%
Fannin	Georgia	-6.7	25,742	$57,073	13.7%	2.9%	3.0%
Taliaferro	Georgia	-6.1	1,558	$52,500	20.4%	2.9%	5.8%
Brown	Indiana	-5.4	15,606	$78,528	7.8%	4.5%	3.5%
Swain	North Carolina	-5.4	14,013	$52,349	19.1%	3.8%	6.6%
Taylor	Georgia	-5.2	7,786	$41,788	32.1%	4.2%	15.5%
Greene	Alabama	-4.8	7,424	$29,200	40.1%	4.5%	13.3%
Watauga	North Carolina	-4.5	55,123	$51,693	25.5%	2.3%	4.4%
White	Georgia	-3.6	28,810	$70,666	12.4%	3.2%	2.6%
Gilmer	Georgia	-3.4	32,426	$74,499	16.0%	3.2%	2.9%
Tishomingo	Mississippi	-3.3	18,660	$53,040	18.0%	2.7%	4.8%
Morgan	Indiana	-3.3	72,659	$79,429	9.3%	3.2%	3.6%

Demographic Profile

Counties with Most Increased Population Density on 2026-02-12

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Issaquena	Mississippi	59.7	928	$31,429	16.2%	2.7%	14.0%
Fairfield	South Carolina	45.2	20,550	$47,885	15.6%	3.2%	5.9%
Osage	Missouri	12.6	13,402	$76,681	8.6%	2.6%	5.6%
Perry	Alabama	12.6	8,031	$37,654	31.4%	3.3%	8.0%
New Madrid	Missouri	10.6	15,731	$51,881	17.7%	3.3%	7.2%
Norton	Virginia	10.1	3,577	$41,495	26.3%	2.3%	10.7%
Brown	Illinois	9.9	6,322	$67,917	9.2%	3.1%	8.5%
Humphreys	Mississippi	9.5	7,395	$33,731	27.0%	3.1%	13.0%
Coahoma	Mississippi	9.3	20,540	$37,461	35.5%	2.3%	14.9%
East Carroll	Louisiana	8.9	7,072	$33,341	35.2%	1.3%	9.2%
Sumter	Alabama	8.4	11,878	$33,310	26.5%	2.7%	9.5%
Leflore	Mississippi	7.9	27,141	$35,277	31.5%	2.1%	14.7%
Holmes	Mississippi	7.8	16,191	$32,538	36.4%	1.9%	15.9%
Lee	Arkansas	7.6	8,359	$34,375	39.4%	2.3%	16.7%
Reynolds	Missouri	7.4	6,010	$45,028	17.3%	3.8%	7.4%

Counties with Most Decreased Population Density on 2026-02-12

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-19.6	99,652	$62,581	13.5%	2.7%	4.1%
Laurens	Georgia	-15.6	49,798	$55,010	22.6%	2.3%	6.3%
Fannin	Georgia	-13.8	25,742	$57,073	13.7%	2.9%	3.0%
Alleghany	North Carolina	-13.8	11,174	$47,172	16.0%	2.6%	6.0%
Clay	West Virginia	-11.3	7,835	$42,318	26.4%	3.5%	10.3%
Gilmer	Georgia	-10.1	32,426	$74,499	16.0%	3.2%	2.9%
Ashe	North Carolina	-9.1	26,950	$56,866	13.5%	3.5%	4.4%
Wilkes	North Carolina	-8.7	65,935	$51,721	16.7%	2.6%	5.8%
White	Georgia	-8.6	28,810	$70,666	12.4%	3.2%	2.6%
Watauga	North Carolina	-8.5	55,123	$51,693	25.5%	2.3%	4.4%
Macon	North Carolina	-8.1	37,941	$59,061	11.6%	3.2%	6.1%
Brown	Indiana	-7.3	15,606	$78,528	7.8%	4.5%	3.5%
Taylor	Georgia	-7.1	7,786	$41,788	32.1%	4.2%	15.5%
Hocking	Ohio	-7.0	27,799	$62,960	16.0%	3.3%	5.7%
Morgan	Indiana	-6.8	72,659	$79,429	9.3%	3.2%	3.6%

Counties with Most Increased Population Density on 2026-01-30

Source: Administrative Regions

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Quitman	Mississippi	11.6	5,774	$32,412	29.6%	2.4%	10.9%
Clarke	Georgia	10.6	129,609	$53,625	26.0%	1.5%	6.3%
Tate	Mississippi	9.3	28,321	$69,704	16.9%	2.7%	3.0%
Chester	Tennessee	9.1	17,611	$59,341	16.1%	1.2%	3.1%
Tipton	Tennessee	8.1	61,553	$74,127	13.6%	2.4%	5.2%
Oktibbeha	Mississippi	7.4	51,771	$46,695	25.5%	1.7%	5.4%
Fayette	Tennessee	6.5	43,267	$88,456	10.3%	3.5%	5.3%
Union	Kentucky	6.4	13,260	$60,327	18.4%	3.0%	5.6%
Lee	Mississippi	6.2	83,034	$67,863	14.6%	2.1%	6.0%
Chicot	Arkansas	5.9	9,765	$39,045	22.6%	3.6%	10.1%
Fulton	Kentucky	5.8	6,409	$38,217	22.6%	2.5%	17.0%
Morehouse	Louisiana	5.6	24,551	$39,454	29.3%	3.6%	11.6%
Athens	Ohio	5.4	61,737	$56,001	21.5%	1.7%	6.8%
Weakley	Tennessee	5.0	33,016	$51,880	19.0%	2.6%	5.5%
Monroe	Arkansas	4.9	6,589	$43,214	20.6%	3.1%	14.5%

Counties with Most Decreased Population Density on 2026-01-30

Source: Administrative Regions

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-28.3	99,652	$62,581	13.5%	2.7%	4.1%
Sharkey	Mississippi	-20.3	3,872	$40,000	30.9%	2.1%	8.0%
Lafayette	Mississippi	-17.5	58,327	$67,185	18.5%	2.0%	6.7%
Tippah	Mississippi	-16.6	21,523	$52,542	18.9%	3.6%	2.2%
Holmes	Mississippi	-15.9	16,191	$32,538	36.4%	1.9%	15.9%
Carroll	Mississippi	-14.7	9,686	$59,327	17.9%	1.7%	4.5%
Pope	Illinois	-13.6	3,739	$60,050	13.3%	4.1%	5.5%
Hancock	Georgia	-13.4	8,650	$40,082	30.6%	3.6%	8.4%
Taliaferro	Georgia	-13.2	1,558	$52,500	20.4%	2.9%	5.8%
Swain	North Carolina	-12.7	14,013	$52,349	19.1%	3.8%	6.6%
Fannin	Georgia	-12.4	25,742	$57,073	13.7%	2.9%	3.0%
Alleghany	North Carolina	-12.1	11,174	$47,172	16.0%	2.6%	6.0%
Tishomingo	Mississippi	-11.7	18,660	$53,040	18.0%	2.7%	4.8%
Humphreys	Mississippi	-10.8	7,395	$33,731	27.0%	3.1%	13.0%
White	Georgia	-10.0	28,810	$70,666	12.4%	3.2%	2.6%

Counties with Most Increased Population Density on 2026-02-12

Source: Administrative Regions

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Lafayette	Mississippi	21.4	58,327	$67,185	18.5%	2.0%	6.7%
Richland	Louisiana	16.7	19,822	$53,544	24.0%	2.9%	8.7%
Oktibbeha	Mississippi	12.2	51,771	$46,695	25.5%	1.7%	5.4%
Norton	Virginia	9.7	3,577	$41,495	26.3%	2.3%	10.7%
Clarke	Georgia	9.5	129,609	$53,625	26.0%	1.5%	6.3%
Rowan	Kentucky	8.2	24,578	$54,321	24.2%	1.3%	8.4%
Davidson	Tennessee	6.9	715,388	$77,853	13.9%	1.9%	6.5%
Athens	Ohio	6.5	61,737	$56,001	21.5%	1.7%	6.8%
Grenada	Mississippi	5.5	21,217	$48,804	24.6%	2.3%	7.9%
Panola	Mississippi	5.2	32,808	$45,945	23.3%	2.1%	6.4%
Osage	Missouri	5.1	13,402	$76,681	8.6%	2.6%	5.6%
Richland	South Carolina	4.9	422,117	$63,784	17.2%	2.2%	7.2%
Christian	Kentucky	4.9	72,069	$55,494	17.1%	1.8%	7.8%
Fulton	Georgia	4.9	1,076,561	$95,292	12.3%	1.8%	10.0%
Fayette	Kentucky	4.8	323,725	$69,479	14.9%	2.2%	7.2%

Counties with Most Decreased Population Density on 2026-02-12

Source: Administrative Regions

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-22.8	99,652	$62,581	13.5%	2.7%	4.1%
Pope	Illinois	-12.5	3,739	$60,050	13.3%	4.1%	5.5%
Alleghany	North Carolina	-11.3	11,174	$47,172	16.0%	2.6%	6.0%
Fannin	Georgia	-11.3	25,742	$57,073	13.7%	2.9%	3.0%
Calhoun	Illinois	-11.1	4,330	$93,203	6.3%	2.7%	2.7%
Taliaferro	Georgia	-10.6	1,558	$52,500	20.4%	2.9%	5.8%
Brown	Indiana	-9.9	15,606	$78,528	7.8%	4.5%	3.5%
Carroll	Mississippi	-9.0	9,686	$59,327	17.9%	1.7%	4.5%
Gilmer	Georgia	-8.7	32,426	$74,499	16.0%	3.2%	2.9%
Ashe	North Carolina	-8.7	26,950	$56,866	13.5%	3.5%	4.4%
Hancock	Georgia	-8.4	8,650	$40,082	30.6%	3.6%	8.4%
Talbot	Georgia	-8.2	5,742	$41,597	25.2%	4.0%	9.6%
White	Georgia	-8.1	28,810	$70,666	12.4%	3.2%	2.6%
Noble	Ohio	-7.9	14,282	$56,098	10.6%	3.3%	7.7%
Harris	Georgia	-7.4	36,086	$97,302	9.0%	2.8%	1.4%